MAILLARD Benjamin

BUT2 Applied Physics
Specialization in Instrumentation Technics

Development of a user interface for a laboratory lysimeter system



Internship carried out at the Forschungzentrum Jülich, Germany
From April 13th to July 3rd, 2026

Supervisor : Nina SIEBERS
Referent teacher : Aurore BRUT
Student : MAILLARD Benjamin

Abstract

During my second year of BUT (University Bachelor of Technologies) Applied Physics, I have carried out my internship at the Forschungszentrum Jülich, Germany. The objective of this internship was to design and implement a graphical user interface for the laboratory lysimeter system, enabling simplified control of key functions such as water level regulation, measurement scheduling and sensor calibration.

I therefore developed a user interface application in LabVIEW, which communicates via a dedicated instrument driver with the DataTaker DT80 data logger that controls the system. This app has therefore been developed to allow users to control the water level, measurement frequency and sensor calibration through an intuitive graphical interface.

Before the system can be used for future lysimeter experiments, the remaining hardware issues related to water level regulation should be resolved and the complete setup should be validated under experimental conditions.

Keywords : Instrumentation, LabVIEW, User interface programming, Environnement

Résumé

Dans le cadre de ma deuxième année de BUT (Bachelor Universitaire de Technologie) Mesures Physiques, j'ai réalisé un stage de 12 semaines au Forschungszentrum Jülich, un centre de recherche situé à côté de la ville de Jülich en Allemagne, au sein de l'IBG-3 Agrosphäre (Institut de Bio et Géosciences, Agro sphère).

Mon objectif principal était de faciliter l'utilisation et de vérifier le bon fonctionnement d'un système lysimétrique à échelle de laboratoire. Pour cela, j'ai programmé et développé une application d'interface utilisateur sur LabVIEW, permettant de contrôler le niveau d'eau présent dans les lysimètres, le temps entre deux mesures et la calibration des nombreux capteurs du système.

Ce stage m'as permis d'acquérir une expérience concrète en instrumentation de systèmes, en programmation d'interface et une capacité d'anticipation de possibles erreurs. J'ai également eu l'opportunité de renforcer mon autonomie et ma rigueur dans un environnement de recherche.

Mots clés : Instrumentation, LabVIEW, Programmation d'interface utilisateur, Environnement

Acknowledgments

First, I would like to thank Dr. Nina Siebers, head of the research group Colloids and Biogeochemical Interface and researcher at the Institute of Bio and Geoscience Agrosphere (IBG-3), who was also my internship supervisor, for her warm welcome to the Forschungszentrum Jülich and for integrating us into the research group.

I would also like to thank Dr. Bei Wu, Dr. Lutz Weihermüller, Dr. Alexander Graf, Dr. Anne E. Berns, Normen Hermes, Shirin Bagheri, Ferdinand Engels, and Niklas Olbrich for their warm welcome at IBG-3 and their aid throughout the internship.

I also want to thank Aurore Brut, my tutor professor from France and the Applied Physics department's international relations coordinator, who offered me this opportunity for an internship abroad and put me in contact with Nina.

Finally, I would like to express my deepest gratitude to Clara Dennery, the other student from the University Institute of Technology (IUT) who worked on this internship with me and with whom I shared this three-month adventure.

Table of Contents

Introduction 6

1. Presentation 7

1.1. Forschungszentrum Jülich 7

1.2. Institute of Bio- and Geosciences, Agrosphere 7

1.3. Context of the internship 8

1.4. Presentation of the system 8

2. Documentation 9

2.1. The Data Logger 9

2.2. LabVIEW communication driver 11

3. User Interface development 12

3.1. The main program 12

3.1.1. Initialization 12

3.1.2. Water Level Control 13

3.1.3. Measurement Frequency Control 14

3.1.4. Sensors Calibration 15

3.2. Log file generator 16

3.3. Timing security 18

4. File Transfer Protocol Server 20

5. Presentation of the User Interface 21

5.1. At start 21

5.2. After unlocking 22

5.2.1. General Information 23

5.2.2. Water Level and Measurement Frequency control 23

5.2.3. Sensors Calibration 24

5.3. Compiling the program into an executable file 24

Conclusion 25

References 26

Glossary 26

Appendix 1

Table of Figures

Figure 1 : sky view of the Forschungszentrum. 7
Figure 2 : System diagram. 9
Figure 3 : DataTaker DT80M-Series 4..... 9
Figure 4 : Diagram explaining the lysimeter monitoring Job. 11
Figure 5 : Communication program between LabVIEW and DT80..... 11
Figure 6 : Schematic diagram of the main state machine. 12
Figure 7 : Sub VI "Send a Water Level Command”. 13
Figure 8 : Start the water level correction. 13
Figure 9 : Read until the end message. 14
Figure 10 : Display of the current real water level..... 14
Figure 11 : Program for changing the measurement frequency. 15
Figure 12 : span coordinates..... 15
Figure 13 : Redox and water level sensors calibration on LabVIEW. 16
Figure 14 : Water Level change log generation. 17
Figure 15 : Calculates the trigger time for the hours. 19
Figure 16 : Calculates the trigger time for minutes. 19
Figure 17 : dEX interface for adding an FTP Server[3] 20
Figure 18 : User interface at the start. 21
Figure 19 : Other sheets of the user interface at start. 21
Figure 20 : User interface after unlocking. 22
Figure 21 : Other sheets of the user interface after unlocking. 22
Figure 22 : General information part of the user interface. 23
Figure 23 : Water level and measurement frequency control part of the user interface. 23
Figure 24 : Redox probs and water level sensors part of the user interface. 24
Figure 25 : Lysimeter controller application executable file 24
Figure 26 : Lysimeter controller application installer folder 24

Table 1: Schedule nomenclature 14
Table 2: Generated log file 18

Introduction

As part of my second year of the University Bachelor of Technology (BUT) of Applied Physics, I completed a 12-week internship at the Forschungszentrum Jülich, a research center in Jülich, Germany. Under the direction of Dr. Nina Sisbers, I worked in the Institute of Bio and Geosciences, Agrosphere (IBG-3).

My main goals for this internship were to experience a new work environment, and I also wanted to discover a new culture, that's why I chose to do my internship in a research center in Germany.

During these three months, I collaborated with my classmate and friend Clara Dennery, from the Toulouse Institute of Technology in France, to complete the project required by Dr. Nina Sisbers.

This project followed the relocation and recommissioning of a lab-scale lysimeter. To help Nina use the lysimeter for future research projects, our main tasks were to verify that the system was working properly, to develop a user interface to simplify its control, and to develop a database and a graphical interface for better management of the lysimeter's measurement data.

In this report, I will first introduce the lysimeter system that I had to work with. Then I will present the user interface program I have developed on LabVIEW, as well as the programming work conducted on the system itself, and finally I will present the finished user interface.

1. Presentation

1.1. Forschungszentrum Jülich

The Forschungszentrum Jülich (Figure 1) is one of Germany's largest interdisciplinary research centers, located in North Rhine-Westphalia, near Cologne. Founded in 1956, it is part of the Helmholtz Association, which comprises 19 German research centers focused on scientific challenges. The Forschungszentrum employs more than 7,000 people from over 70 different nationalities. It specializes in key areas such as energy, environment, and information technology.

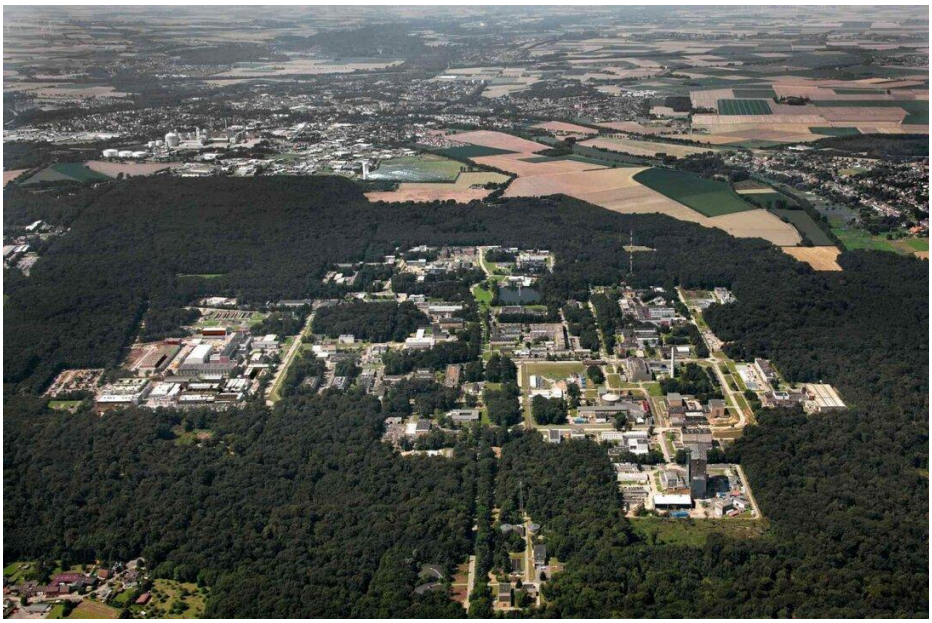


Figure 1 : sky view of the Forschungszentrum.

Jülich is home to unique research facilities, such as supercomputers like JUWELS, one of Europe's highest-performance supercomputers. It plays a significant role in the energy transition, particularly through research on renewable energy, energy storage, and sustainability.

1.2. Institute of Bio- and Geosciences, Agrosphere

During my internship, I worked in the IBG-3 (Institute of Bio- and Geosciences, Agrosphere) department of the Forschungszentrum Jülich, which studies terrestrial ecosystems and their interactions with the atmosphere, the biosphere, and the soil. Its research focuses in particular on the study of water, carbon, and nutrient cycles, and the impact of climate change and agricultural practices on soils and crops. Through experimental and modeling approaches, IBG-3 contributes to the development of sustainable strategies for optimized natural resource management and agriculture that is resilient to environmental challenges.

1.3. Context of the internship

For research on the phosphorus cycle, Dr. Nina Siebers, my internship supervisor, would like to use the lysimeter system that she received as a donation from Prof. Dr. Peter Leinweber of the University of Rostock, who had previously used this system to study loss of phosphorus from soils due to alternating redox conditions.

Following this change of ownership, the system was disassembled for transport to the Forschungszentrum Jülich and then reassembled.

Our tasks during the internship, with my classmate Clara, with whom I'm working on this project, were to verify that the system is working correctly after reassembly, to simplify its use by developing a user interface to control the various configuration parameters, and to develop a database and a graphical interface to simplify data processing. I personally focused on how the system works and the development of the user interface, while Clara worked on the database and the graphical interface.

1.4. Presentation of the system

The system that I worked with is composed of three lysimeters. A lysimeter is a scientific device used in agronomy to study water exchange between the soil, plants and the atmosphere. It is a cylinder into which a soil sample is placed, with several sensors inserted into the soil to measure data. The cylinder part is called a monolith. These devices can range in size from 10 cm to 5 m in diameter and from 20 cm to 3 m in depth.

Here, each of the three lysimeters I had worked with (Figure 2) was made from a PVC pipe with an internal diameter of 30 cm and a length of 90 cm. Next to each monolith is a column that can hold water, making it possible to observe the water level inside the monolith. The water level in the lysimeters is kept constant by water level sensors and an automatic valve and pump system. In each monolith, three combined pH and redox probes, three tensiometers, and three soil temperature and moisture sensors are installed, with one of each type at heights of 25 cm, 50 cm, and 75 cm, allowing observation of the different soil layers.

A DataTaker DT80 data logger served as the central control unit of the system. It triggered measurements from the connected sensors, stored the acquired data and managed the regulation of the water level.

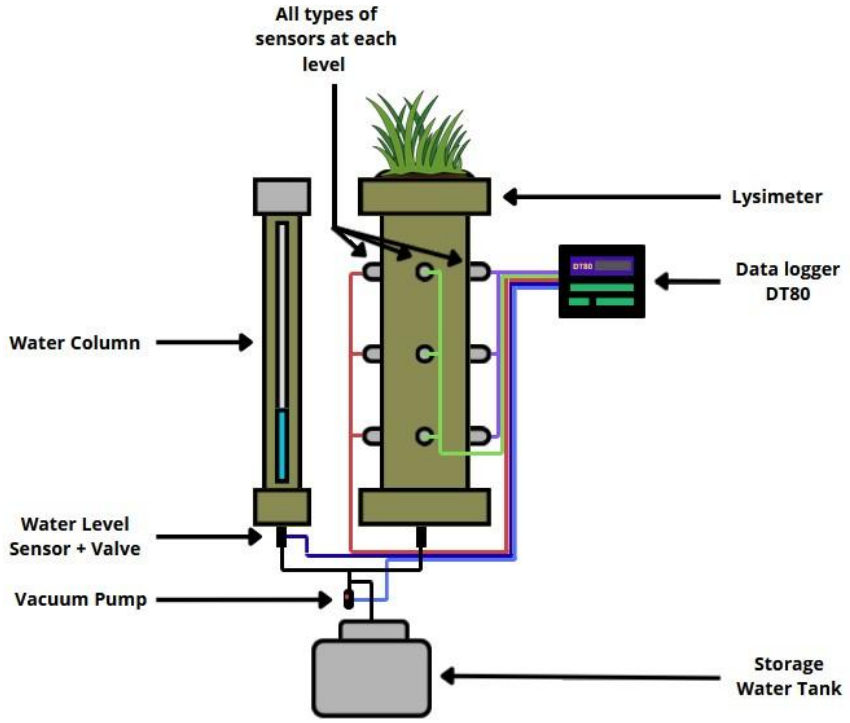


Figure 2 : System diagram.

2. Documentation

At the beginning of the internship, it was necessary to become familiar with the structure and operating principles of the lysimeter system. For this purpose, Nina provided a technical documentation folder containing datasheets and manuals for the individual system components, as well as documentation from previous research carried out with the setup. In addition, Niklas, a technician at the institute who had already worked on commissioning the lysimeters, explained the main hardware components, control processes and operational issues of the system.

2.1. The Data Logger

A large part of understanding the system was focused on the data logger. A dataTaker DT80M-Series 4 (Figure 3) manufactured by Thermo Fisher company.



Figure 3 : DataTaker DT80M-Series 4

This data logger can record measurement data and has a large storage capacity. It can communicate with a computer via a modem, a USB connection or an Ethernet connection. It has 5 analog ports and 4 digital ports, allowing the use of a wide range of sensors. This data logger also features a software interface, named “dEX” enabling real-time data monitoring and viewing.

The data logger has an Ethernet connection, we quickly decided, with Nina, to connect it to the institute’s intranet, so everyone can have access to it from their own office, just by having his IP address[1] .

Furthermore, as this data logger is programmed using a specific language developed by the supplier, I had to familiarize myself with this language to understand how the lysimeters work, and the basics of this language were all explained in the DT80 data sheet.

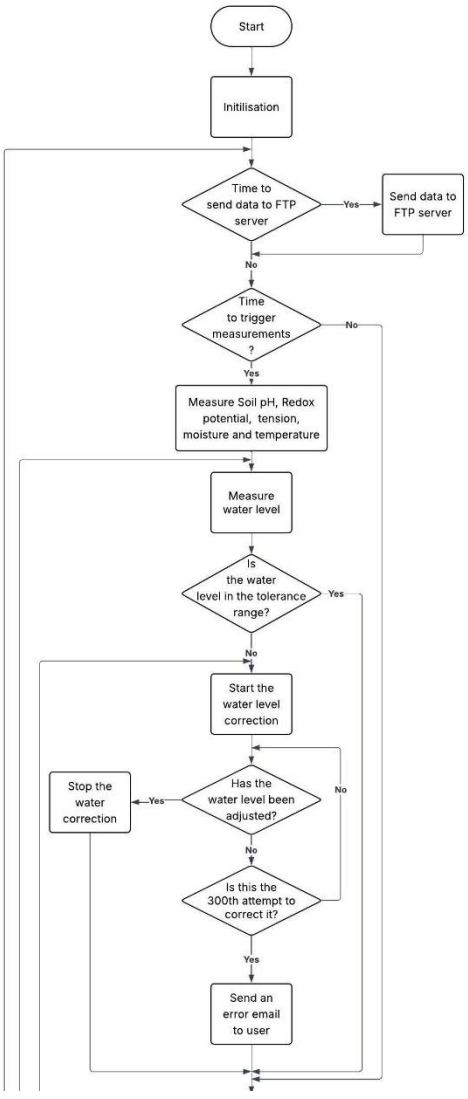
With this language, the program, which is called a “Job”, is separated into distinct functions called “schedules”, and these ones are automatically called after a specified time to execute the desired actions or can also be called manually by a user. To summarize the lysimeter monitoring program (Figure 4):

1. The Job initializes by defining the various variables and conversion scales needed for the rest of the program; this step is done only once.

2. The job checks whether it is time to upload the measurement data to a storage server and initiates the upload when required.

3. The Job checks if the interval between two measurements has been reached. If so, it :
   a. Triggers the sensor measurements.

   b. Checks and adjusts the water necessary.



4. The Job checks whether the manual schedules have been triggered

5. The Job checks whether the end-of-job schedule has been triggered manually

6. The Job repeats until the end-of-job schedule is triggered

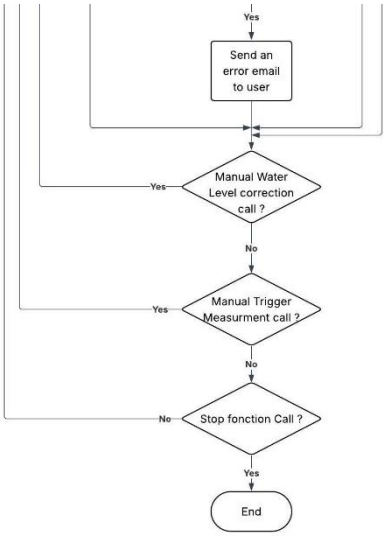


Figure 4 : Diagram explaining the lysimeter monitoring Job.

2.2. LabVIEW communication driver

During my research, I found a communication driver for LabVIEW and the DT80, which allows data to be exchanged between the data logger and a LabVIEW program. As the data logger is connected to the institute's network, all I need to do is open a TCP/IP[2] connection in LabVIEW using the DT80's IP address1 , then send the data or commands to the data logger and finally close the connection (Figure 5).

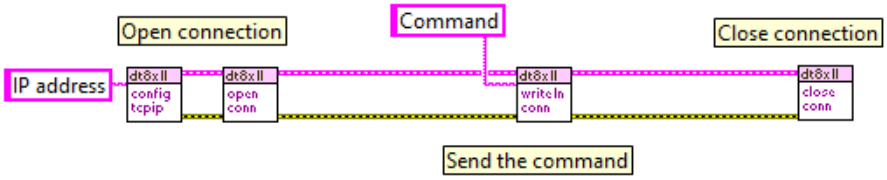


Figure 5 : Communication program between LabVIEW and DT80

As LabVIEW is a graphical programming environment widely used for measurement, automation and control applications, it was well suited for developing the user interface for the lysimeter system, and as I already have some experience using it, I decided, with Nina's approval, to develop the user interface application in LabVIEW. The communication driver was very useful for this, simplifying all communication between LabVIEW and DT80.

Developing the user interface in LabVIEW also allows the DT80 Job to run without keeping the user interface program running, meaning a computer does not need to be dedicated to the lysimeters.

3. User Interface development

The user interface was intended to provide access to the main control functions of the lysimeter system, including water level regulation for each lysimeter, configuration of the measurement interval, and sensor calibration. Previously, the previous owners of the device had to contact UGT, the system's manufacturer, and pay to have any changes made to the system's settings, this user interface will therefore also enable users to operate independently of UGT.

3.1. The main program

The LabVIEW application was structured as a state machine (Figure 6). This programming architecture allows user actions, such as pressing a button, to trigger specific states instead of executing all functions simultaneously.



Figure 6 : Schematic diagram of the main state machine.

This state machine starts with an initialization, and then waits for any event to happen ; here, an event is a button being pressed to trigger a change. If a button is pressed, it does the change and then waits for another event. If the "Quit" button is pressed, the program ends and stops.

3.1.1. Initialization

To start the user interface program, I wanted to force the user to do the first three actions:

- Enter the IP address1 of the DT80.
- Enter the username.
- And enter a file to save the log data.

So, the program disables and greys out all the elements the user can interact with until all three have been entered.

After the user enters these three elements, the program tries to start communication with the DT80 by sending a command to it. This makes it possible to check if the DT80's IP address1 is correct or not. If the IP address1 is incorrect, an error will occur and will be added to the LabVIEW error cluster, which allows the program to check if the IP address[1] is correct. If it's correct, the LabVIEW program gets all the data it needs at start from the DT80's RAM[4] and enables the use of the buttons. If the IP address[1]

is wrong, the program will reset the IP address[1] and ask for a new one to repeat the initialization.

3.1.2. Water Level Control

To control the water level, we need to change the value of the variable in the DT80's RAM[4] that corresponds to the water level. For example, to control the water level of the first lysimeter, you need to change the value of 1CV, the variable that stores the value of the first water level, you send the command "1CV=X" where X is the water level of the first lysimeter that you want in mmH_20. Once the DT80 receives the command, it will respond with "1CV X" to confirm the change.

On LabVIEW, when the button "Set water level" is pressed, the program must create the command from the value on the user interface, send it to the DT80, and after verifying that the response is correct (Figure 7).

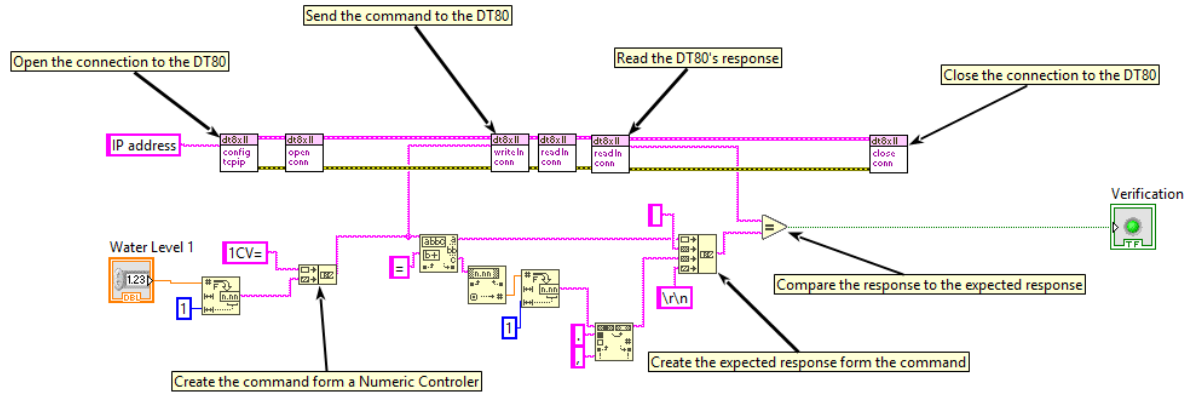


Figure 7 : Sub VI "Send a Water Level Command".

Once the new value has been set in the DT80's RAM[4], we need to send the command "XH", which starts the schedule H, the one that initiates the water level correction with the new value of the variable 1CV (Figure 8).

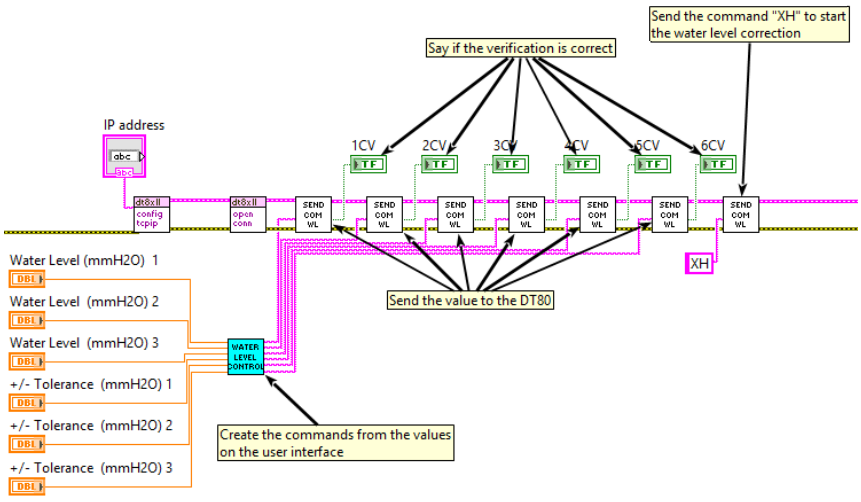


Figure 8 : Start the water level correction.

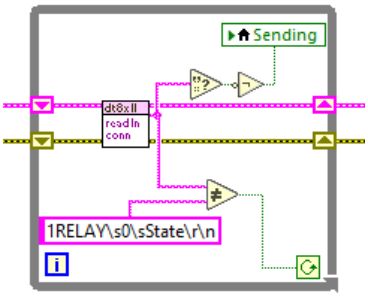


Figure 9 : Read until the end message.

Then, the program will disable and gray out every button until the end message of the water level regulation (Figure 9), to prevent the user from sending further commands, and thus too much information, to the DT80 before it has finished this task.

And finally, to complete the water adjustment, the program requests the current real value to display it on the user interface (Figure 10).

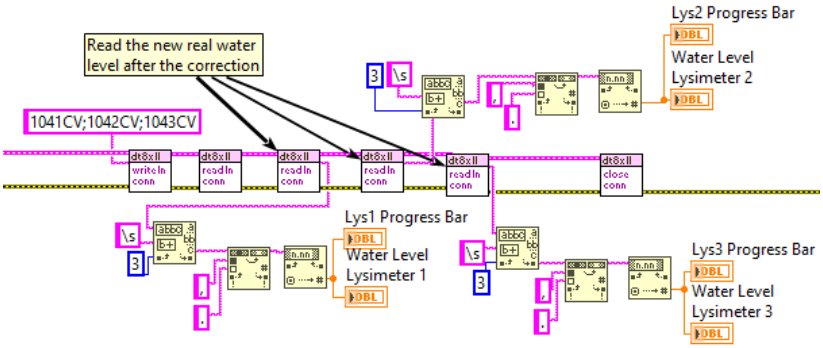


Figure 10 : Display of the current real water level.

3.1.3. Measurement Frequency Control

To control the measurement frequency, meaning the time between two measurements, we must change the time of activation of schedule G, the one for the automatic triggering of measurements. For that, the program has to send to the DT80 the command "RGnY" with "n" a number and "Y" the unit (Table 1)

Trigger	Run every n	Range
nD	Days	1 – 65535
nH	Hours	1 – 65535
nM	Minutes	1 – 65535
nS	Seconds	1 – 65535
nT	Milliseconds (Thousandths of seconds)	5 – 65535

Table 1 : Schedule nomenclature

For the current Job in the DT80, we must limit the measurement interval to at least 5 minutes to prevent the data logger from overheating.

Furthermore, to have all the data available when the LabVIEW application starts, it would be necessary to retrieve the value of this measurement frequency, and to do this, it would need to be stored in the DT80's RAM[4]. But there is no command that can request the trigger time of a schedule, so to have this information, I store it in two variables, 18CV for the number "n" and 19CV for the Units "Y". However, the DT80 can only accept numeric variables, while "Y" must be a letter. So, I assigned a number to each letter, and the LabVIEW program converts them automatically whenever it needs this information (Figure 11).

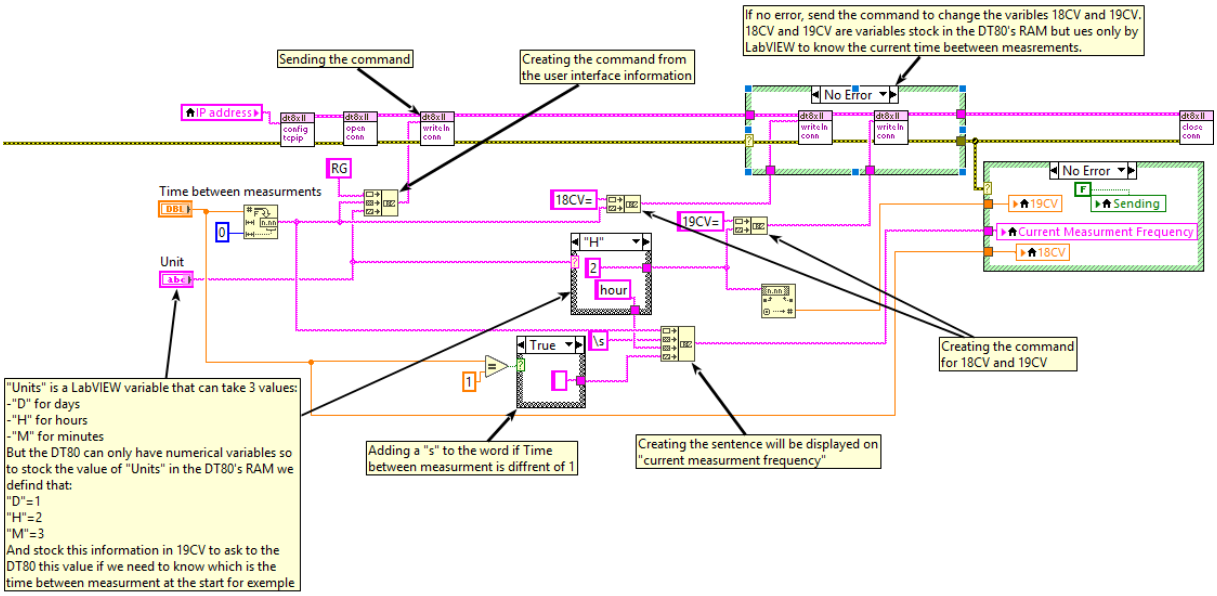


Figure 11 : Program for changing the measurement frequency.

The program is sending the command to change the trigger time of the schedule G, and if there is no error, it will change the value of the two variables, 18CV and 19CV, to keep in memory what is the current time between the measurements and be able to retrieve this information to LabVIEW if needed.

3.1.4. Sensors Calibration

The sensors used on the device use a conversion scale to provide their measurement. All the scale conversions are defined in the initialization of the DT80's Job but can be change and stock, as before, in the DT80's RAM[4].

For this installation, the sensors used employ only two types of conversion scale, linear conversion or polynomial conversion.

The linear conversion scale is also called "span". A span transforms a measured signal value (e.g., mV) into the corresponding physical value (e.g., kPa) using a straight-line function. The syntax of a span is Sn=a,b,c,d with "n" a number and "Sn" the name of the span. a and b are the physical coordinates of two points on the calibration line, and c and d are the signal coordinates of two points on the calibration line (Figure 12).

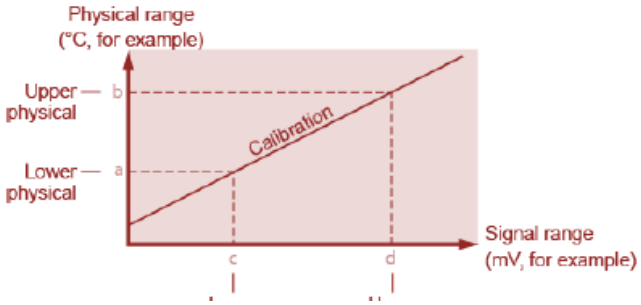


Figure 12 : span coordinates

The polynomial scale conversion uses a polynomial to convert the signal measured in a physical value. The syntax of the polynomial is Yn=k_0,k_1,k_2,k_3,k_4,k_5 with "n" a number and "Yn" the name of the polynomial scale. The data logger uses, for the conversion, a polynomial of the form:

$$Y = k_0 + k_1 \times x + k_2 \times x^2 + k_3 \times x^3 + k_4 \times x^4 + k_5 \times x^5$$

So, to change the calibration of the sensors, you must change the scale conversion of the corresponding sensor. For that you can send a command of the form Sn=a,b,c,d or Yn=k_0,k_1,k_2,k_3,k_4,k_5 depending on the sensor.

As before, the program must create the command by some string manipulation, and after sending it to the data logger (Figure 13), the DT80 Job will automatically use the new sensor calibration for the next measurement.

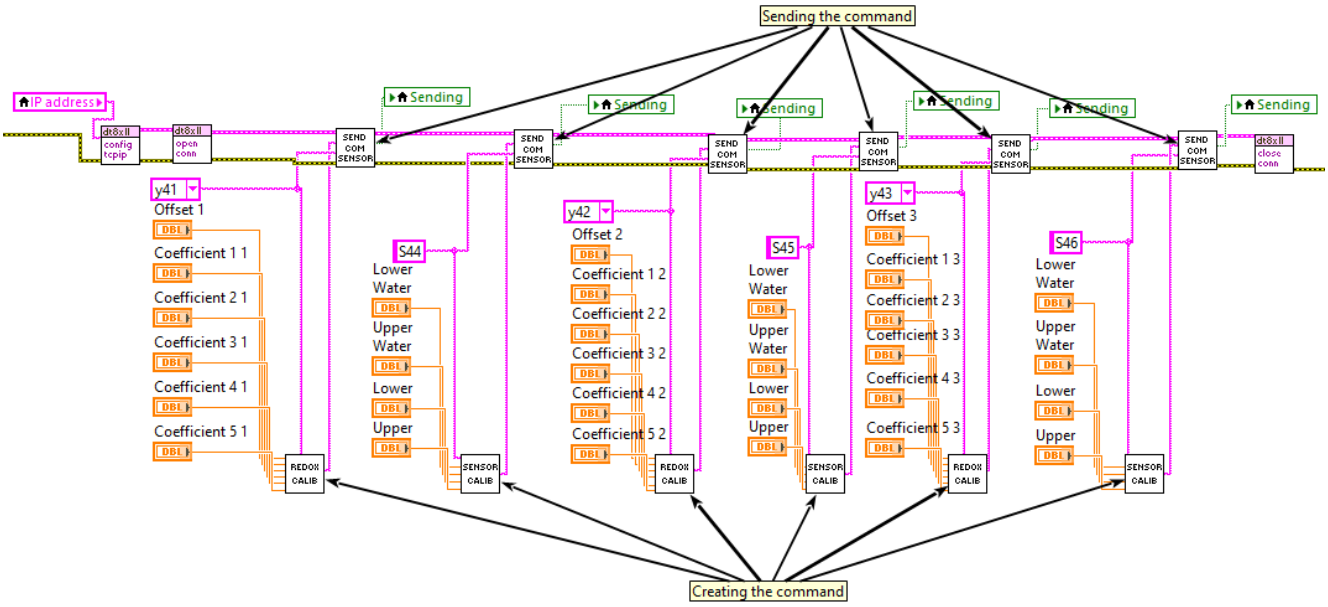


Figure 13 : Redox and water level sensors calibration on LabVIEW.

3.2. Log file generator

To improve the traceability of system operation, Nina requested the implementation of a log file that records and stores any changes made to the system. For that, the user must enter their username and the log file path at the start of the program, and to make sure that the user did it, everything else is disabled and grayed out until it's done. Once the interface has been unlocked, every user-initiated change is applied by the program and simultaneously recorded in the log file. The corresponding log entry is then appended to the Excel file selected by the user at startup.

The saved sentence contains the date and the time when the change was made, the username of the user who made the change, what was changed, and the old and new values of the change.

First, the program takes the old values that are stored in a cluster and uses them to create a part of the log sentence, then takes the new values that are currently on the user interface and writes the rest of the log sentence. Finally, it overwrites the old value with the new value in the cluster, to make the new value become the old value for the next time there is a change (Figure 14).

After making the sentence, the program opens the file defined by the user, writes the line at the end of the file and closes it.

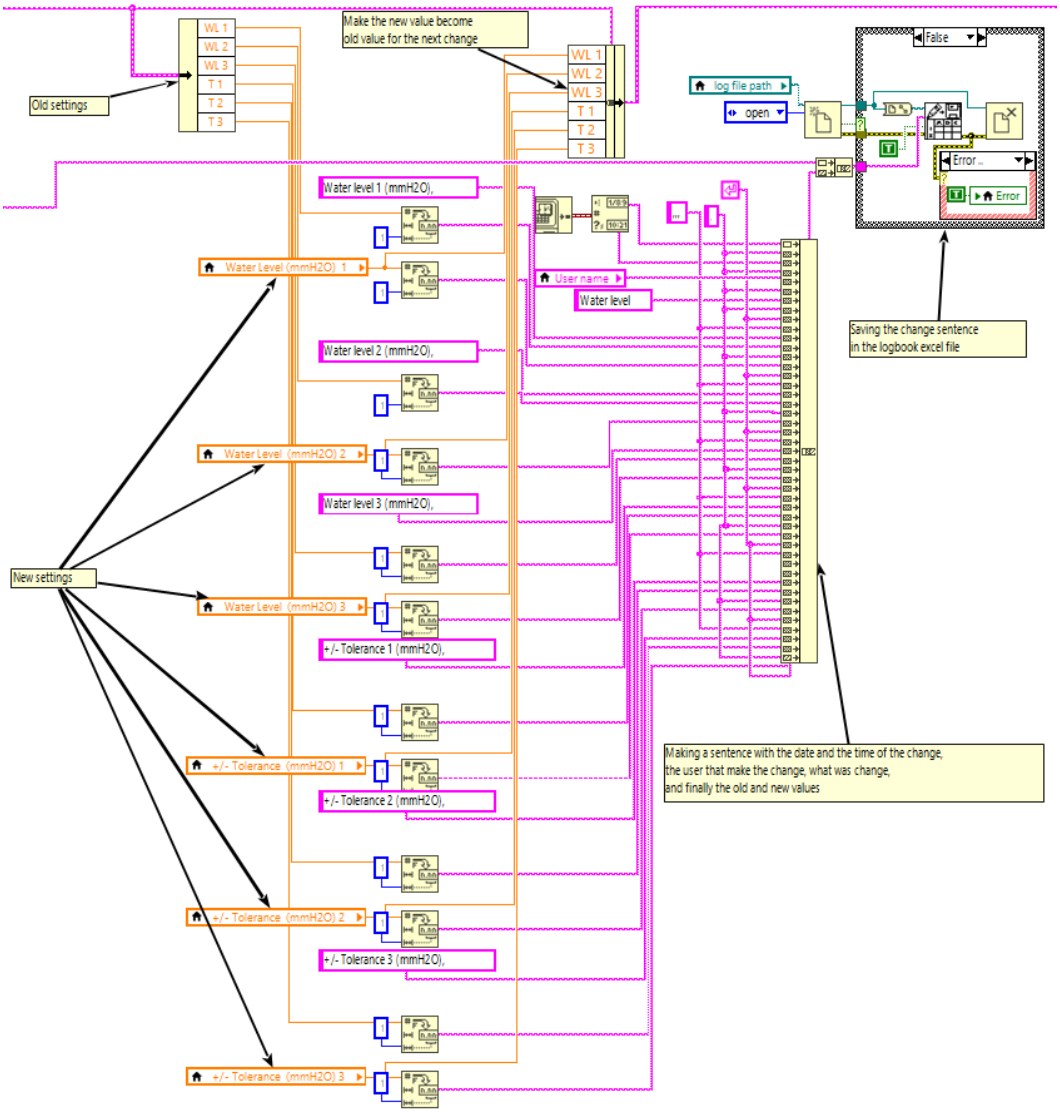


Figure 14 : Water Level change log generation.

After the generation, the log is added to the Excel file (Table 2)

This allows the user to analyze the data by comparing the evolution in response to changes in parameters, as they know exactly when the change took place.

Date	Time	User name	Value changed	Last value	New value
6/2/2026	12:18 PM	Benjamin	Measurment fréquency	1 Hour	2 Hours
6/2/2026	12:18 PM	Benjamin	Water level		
			Water level 1 (mmH2O)	200	200
			Water level 2 (mmH2O)	100	150
			Water level 3 (mmH2O)	100	100
			+/- Tolerance 1 (mmH2O)	5	5
			+/- Tolerance 2 (mmH2O)	5	5
			+/- Tolerance 3 (mmH2O)	5	5
6/2/2026	12:18 PM	Benjamin	pH sensors calibration		
			pH 1.1		7.0;4.0;-20.0;144.0
			pH 1.2		7.0;4.0;-4.0;129.0
			pH 1.3		7.0;4.0;26.0;118.0
			pH 2.1		7.0;4.0;2.5;175.0
			pH 2.2		7.0;4.0;16.0;185.0
			pH 2.3		7.0;4.0;6.0;175.0
			pH 3.1		7.0;4.0;-10.4;164.0
			pH 3.2		7.0;4.0;11.6;177.0
			pH 3.3		7.0;4.0;52.0;127.0
6/2/2026	12:18 PM	Benjamin	Tensiometer sensors calibration		
			Tensiometer 1.1		-20.0;100.0;100.0;1000.0
			Tensiometer 1.2		-20.0;100.0;100.0;1000.0
			Tensiometer 1.3		-20.0;100.0;100.0;1000.0
			Tensiometer 2.1		-20.0;100.0;100.0;1000.0
			Tensiometer 2.2		-20.0;100.0;100.0;1000.0
			Tensiometer 2.3		-20.0;175.0;100.0;1000.0
			Tensiometer 3.1		-20.0;100.0;100.0;1000.0

Table 2 : Generated log file

3.3. Timing security

After some tests, I saw that it can be a problem to send a command to the DT80 while another task is running ; sometimes the data logger loses some information and does not apply the change. So, to make sure that the user does not have any problem with the DT80 by using the user interface, I had to improve the security of the LabVIEW program.

I had already disabled and greyed out all buttons whenever the user initiated a command on the DT80. However, there was still another situation in which the data logger was busy without direct user interaction: when the DT80 automatically triggered a measurement.

Triggering the measurement can take 2 to 5 minutes, and the user should not be able to trigger a change at this time to make sure no any problem occurs.

For that, I had to synchronize the LabVIEW program to the DT80's Job. It is written in the DT80 documentation that the data logger is referenced to midnight. This means that if you change the measurement interval to 1 hour at 2.37 pm, for example, the DT80 will take measurements at 3 pm, 4 pm, 5 pm... etc. and not at 3.37 pm, 4.37 pm... etc. Therefore, as long as the program knows the configured measurement interval, it can determine the expected measurement times based on the DT80's midnight reference. The exact time at which the interval was changed does not need to be stored or used for this calculation.

Now, to calculate when it should happen, it depends on the unit of the measurement frequency, but the program knows this unit because the information is stored in the variable 19CV. So, the program first needs to check the variable 19CV to know which unit it has to use to calculate the time to disable the user interface. Then, with the value of the time between the measurement, which is stored in the variable 18CV, the program can calculate when the automatic measurement is supposed to be triggered.

If 19CV=1, it means the unit is a day, the measurement is triggered at midnight on a certain day, depending on the value of 18CV. In that case, there's no need to disable and grey out the buttons.

If 19CV=2, it means the unit is an hour, it is triggered at the transition between two hours, for example, at the transition from 1:59 p.m. to 2:00 p.m. So, the first condition of the trigger is that the minutes are 00. Then, to determine if it is the right time to trigger, the program must divide the current time by the variable 18CV, which represents the measurement frequency value, and if the remainder of the division is 0, then it is the trigger hour. If the two conditions are true, it's the trigger time (Figure 15) and the LabVIEW program have to disable and gray out the button on the user interface.

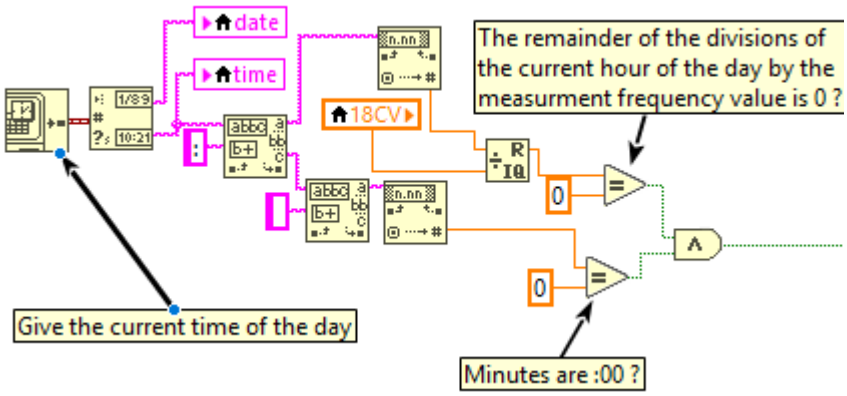
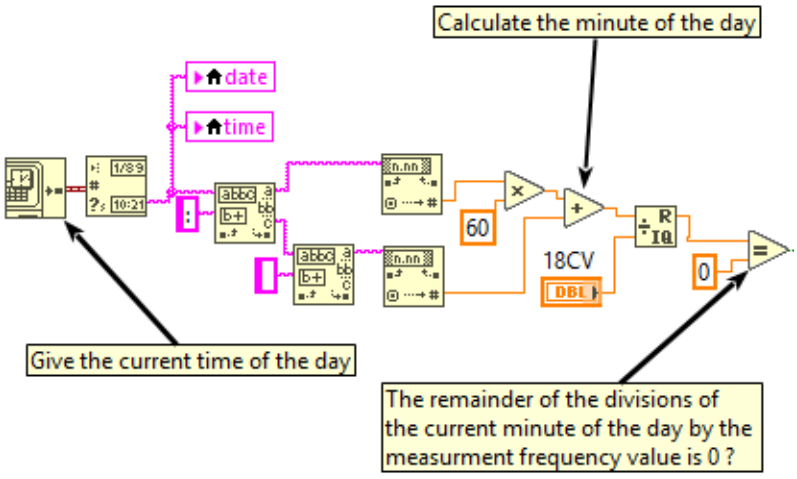


Figure 15 : Calculates the trigger time for the hours.



If 19CV=3, it means the unit is a minute, so the program must calculate the current minute of the day, and then if the reminder of the division of the current minute of the day by 18CV is 0, then it is the trigger time (Figure 16), and the LabVIEW program has to disable and gray out the button on the user interface

Figure 16 : Calculates the trigger time for minutes.

4. File Transfer Protocol Server

The user interface is working. Now, with Clara, we would like to be able to automatically retrieve measurement data from the DT80 so that we can automatically add it to the database she created.

The DT80 can send its data to an FTP Server[3] with a TCP/IP[2] connection, through the institute's network. This is how they retrieved the data before the device was moved to the Forschungszentrum, but now the DT80 cannot have access to the old FTP Server[3] they used before, because it was a private network.

So, we had, with Clara, to create a new FTP Server[3] , in the institute this time, and modify the DT80's Job to change the FTP Server[3] address to finally get the data. While Clara is creating a new FTP Server[3] , I was modifying the DT80's Job.

To modify the DT80's Job, I used "dEX" (Figure 17), the DT80's programing interface. dEX is a WYSIWYG (*What You See Is What You Get*) editor designed for data acquisition. This means that to code with dEX, you simply add an element to the code list, enter the parameters associated with that element, and then dEX automatically generates a line of code and adds it to the Job.

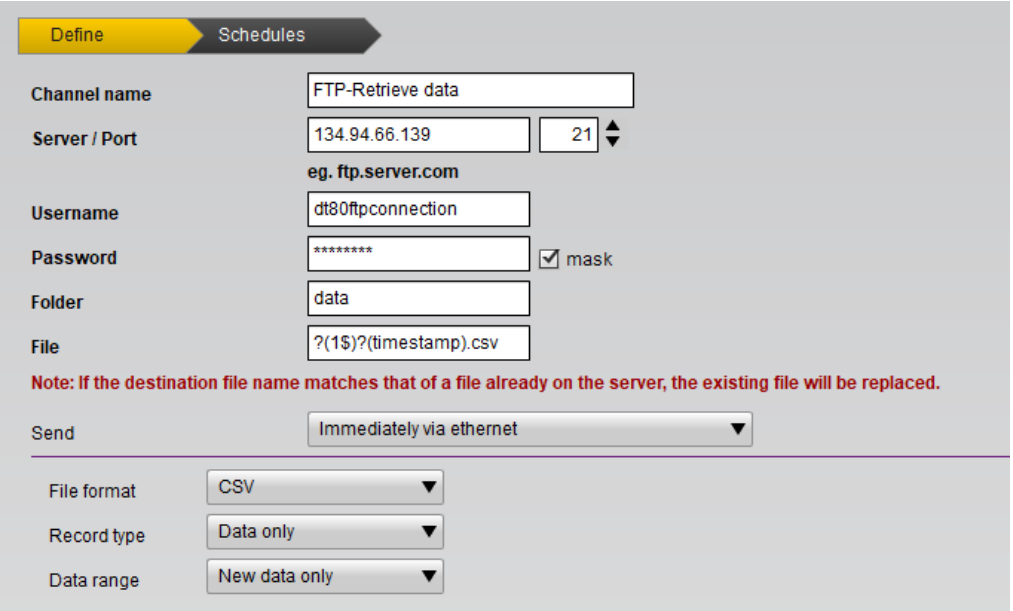


Figure 17 : dEX interface for adding an FTP Server[3] .

To change the FTP Server[3] where the DT80 send his measurement data, I had to change the "Server/Port", the "Username", the "Password", the "Folder" and the "File". When Clara had finished creating and organizing the FTP Server[3] , she gave me all the information to create the connection between the DT80 and the FTP Server[3] . Then I only had to specify how I wanted the file to be sent, here via ethernet connection, what file format for the data retrieve, and which schedule data I wanted. And now the automatic data retrieval is done.

5. Presentation of the User Interface

5.1. At start

When starting of the LabVIEW program, everything is disabled and grayed out (Figure 18, Figure 19), to force the user to enter the IP address[1] , their username and the log file path. Also, the user interface tells the user what he must do to unlock the user interface in the rectangle at the top right corner. This rectangle is telling information to the user throughout the program execution.

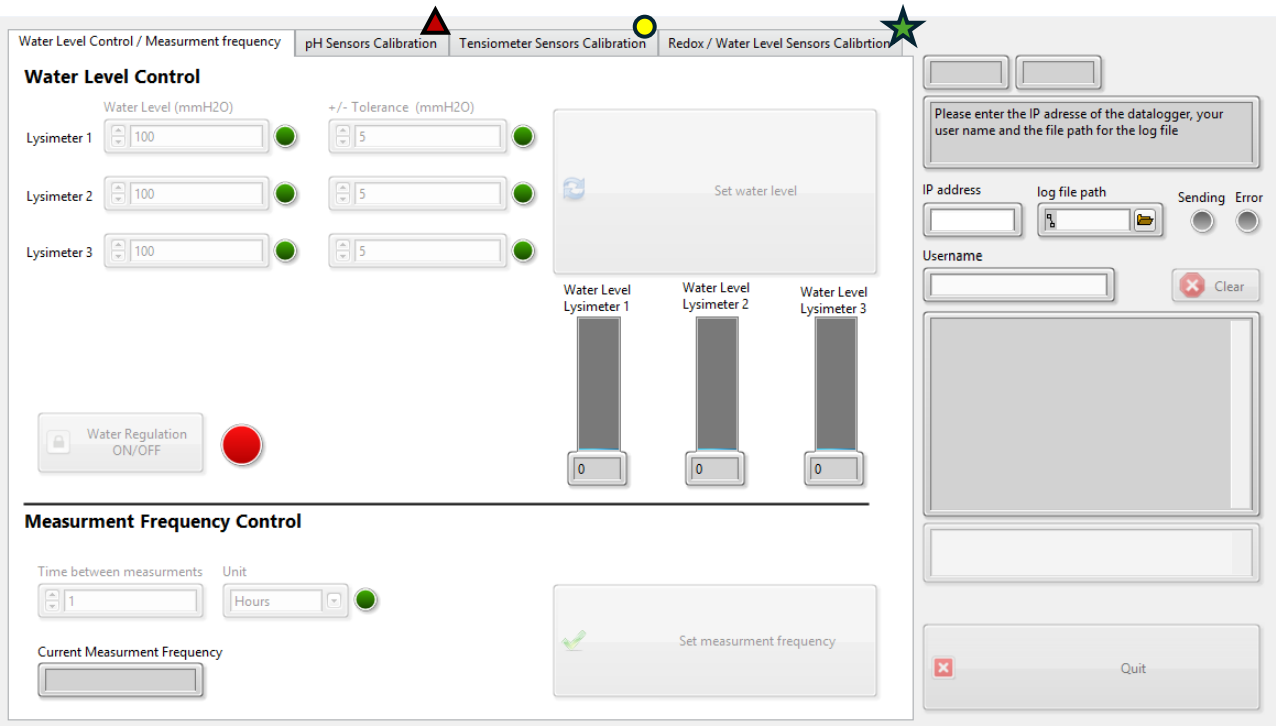


Figure 18 : User interface at the start.

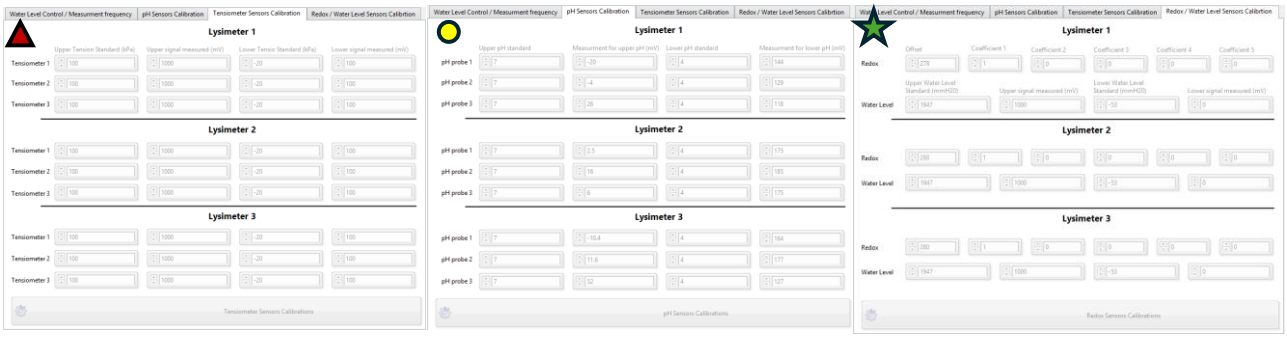


Figure 19 : Other sheets of the user interface at start.

After having entered the three needed elements, the user interface is unlocked.

5.2. After unlocking

After unlocking, all the items of the user interface are available (Figure 20, Figure 21)

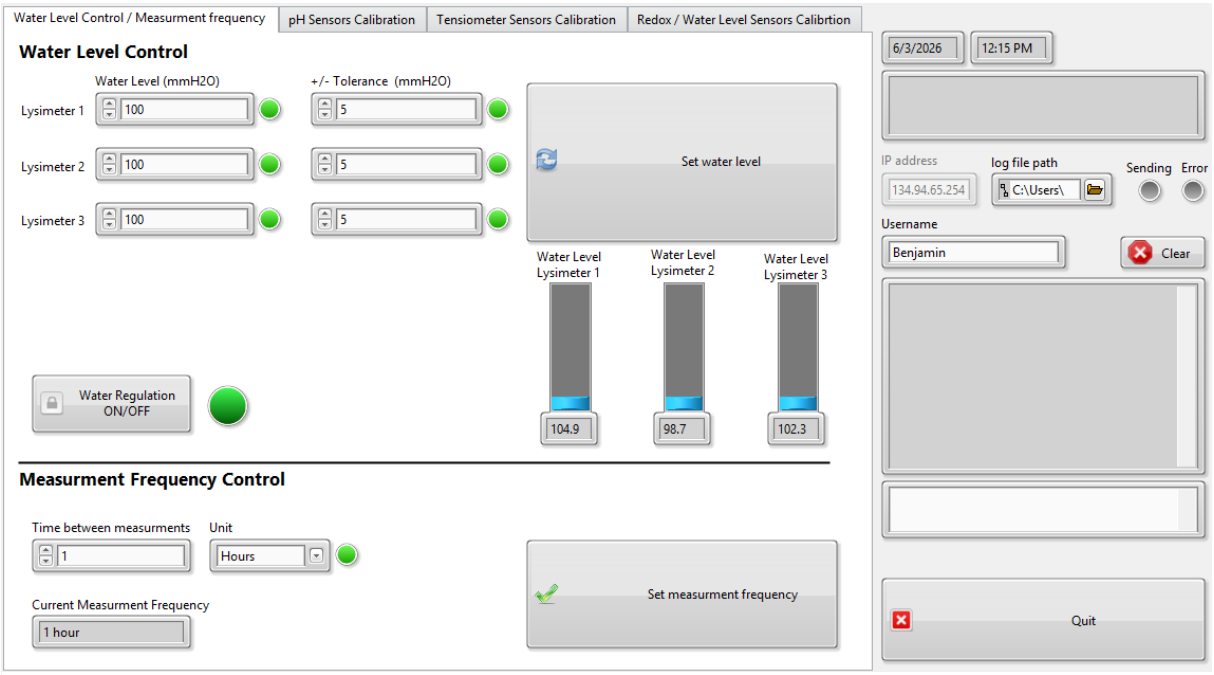


Figure 20 : User interface after unlocking.

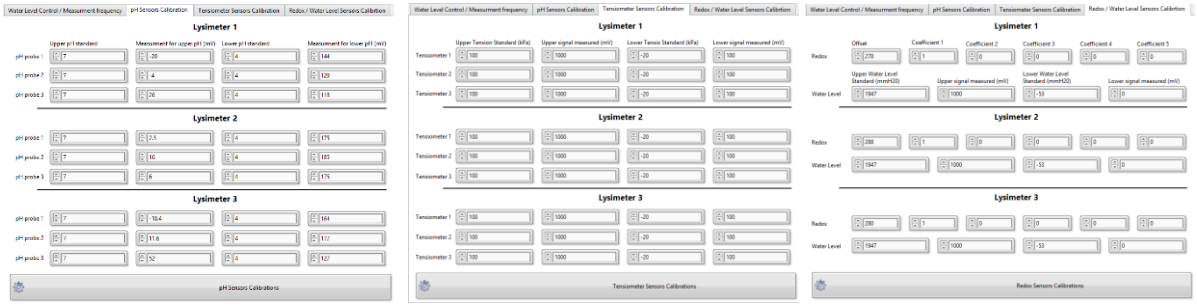


Figure 21 : Other sheets of the user interface after unlocking.

The user interface for the lysimeters control is organized in five parts:

- General information on the right
- Water Level and Measurement Frequency control, the first page of the selection menu
- pH probs calibration, the second page
- Tensiometer sensors calibration, the third page
- And Redox probs and water level sensors, the fourth page

To use this interface, each time the user wants to make a change, he has first to put the value on the corresponding numeric controller and then push the button to send the change to the DT80.

5.2.1. General Information

On the general information part (Figure 22), the user can find a box that gives him information on how the program is running, like whether the change he wants occurred well, if he must wait for a response of the DT80... ect.

Right under, there is the IP address[1] , the username and the log file path that the user must enter to have access to the rest of the user interface.

Next to that, there are two communication LEDs, the “sending” one shows if a communication occurred and the “error” one shows if an error happened. After that the communication to the user box confirms it with a sentence.

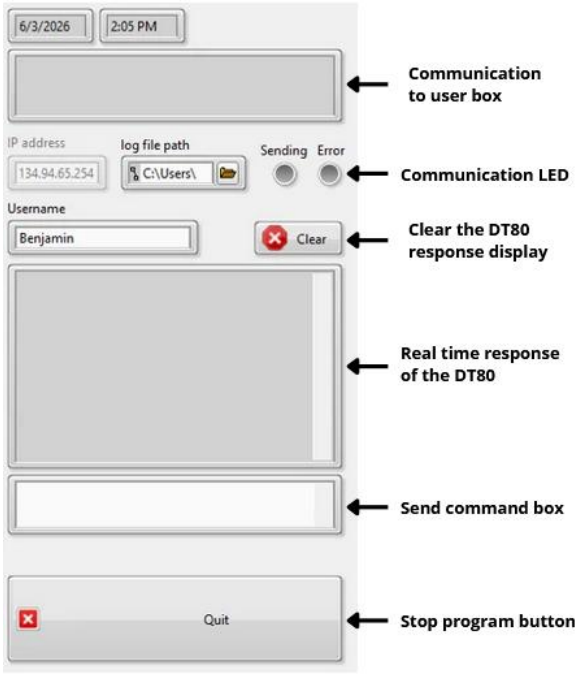
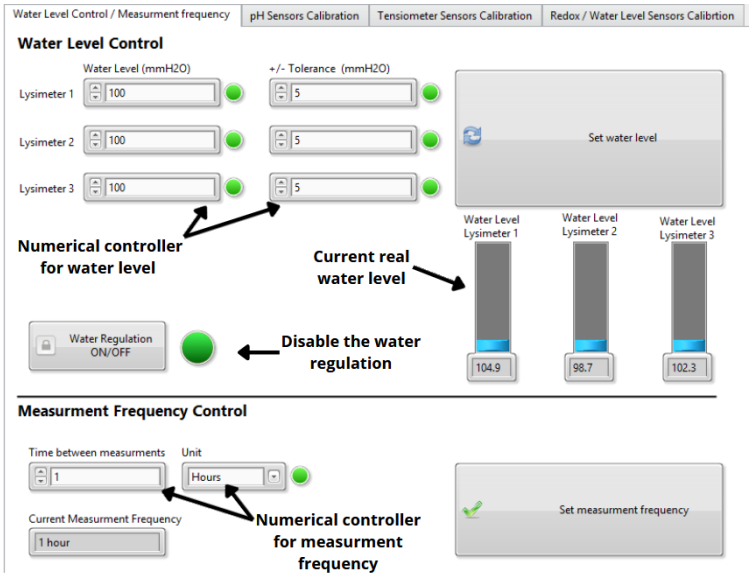


Figure 22 : General information part of the user interface.

Under the LEDs, there is one button and two boxes, which are directly communicating with the DT80. The bigger box reads all the messages that the data logger sends and with the smaller one the user can send a command directly from the interface. There is also a clear button to clear the DT80 response display. This part is approximately the same thing as the communication page on dEX, but directly on the LabVIEW interface.

Under that, there is the “Quit” button to end the use of the user interface.

5.2.2. Water Level and Measurement Frequency control



On the water level control part (Figure 23) there is first all the numerical controller for the water level and the tolerance, and the button to send the command to the DT80. Next to all the numerical controllers, there is an LED that turns on if the value that is displayed on the interface is the same value in DT80’s RAM[4] and so the value is currently used in the DT80’s Job.

Figure 23 : Water level and measurement frequency control part of the user interface.

Under the button, there is the current real water level that is displayed.

I also added a button to disable the water level regulation ; several issues have been encountered, so if the user would prefer not to have it, that is possible.

On the measurement frequency control part, there are two controllers, one for the value and one for the unit, the button to send and change the measurement frequency, and the LED like before with the water level. There is also a display screen that displays the current measurement frequency.

5.2.3. Sensors Calibration

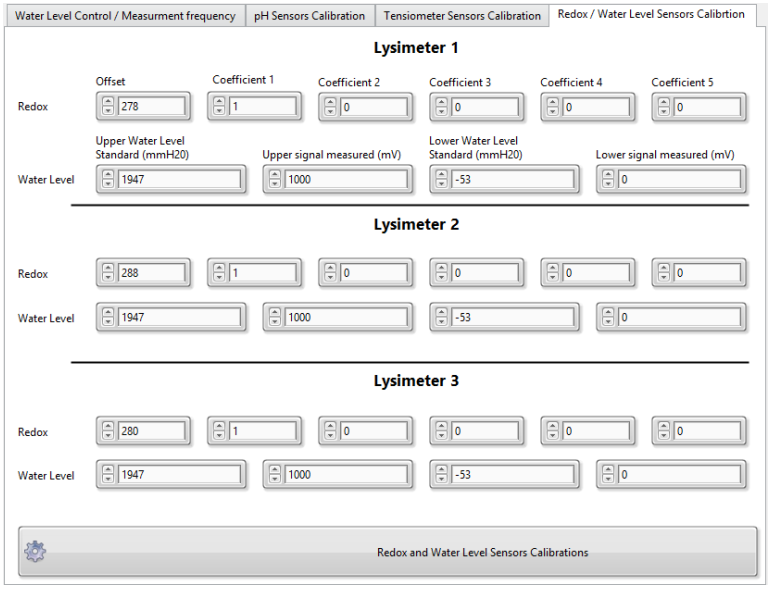


Figure 24 : Redox probs and water level sensors part of the user interface.

The sensor calibration tabs (Figure 24) follow a similar structure. They contain several numerical input fields in which the user can enter the required calibration parameters. A dedicated command button at the bottom of each tab sends the entered values to the DT80 and updates the corresponding sensor calibration settings.

5.3. Compiling the program into an executable file

To finish the user interface, I converted the program into an executable file (.exe) (Figure 25) so that it could be used on any computer without needing to install LabVIEW.



Figure 25 : Lysimeter controller application executable file

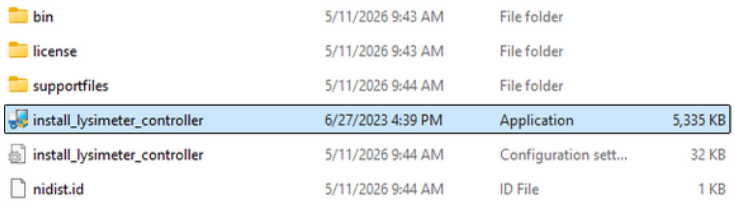


Figure 26 : Lysimeter controller application installer folder

And I also created an installer folder (Figure 26) to install this application on any computer that needs it.

Conclusion

In conclusion, I completed my internship at Forschungszentrum Jülich, ending my second year of study in Applied Physics. During this Internship, I have improved my programming skills, my autonomy, and my ability to adapt to the problems I meet. I had the chance to apply many of the technical skills I acquired during my training in Applied Physics. I also discovered the world of research centers which was new to me.

My main project was to develop a user interface for the control of a lysimeter system, and after some modification and test during the last two weeks of internship, the application will be ready to be used to control the device. Also, during these last weeks, I will write a user guide to use well the application.

By simplifying the control of water regulation, the measurement frequency and the calibration of sensors, this new graphical interface frees IBG-3 users from any dependence on UGT for system monitoring.

Clara, for her part, has developed a database and a graphical interface, simplifying the system's data processing.

There are still several hardware issues on the system concerning the temperature and moisture sensors, and on the water regulation concerning the water level sensors and the pump, but once these issues have been resolved, the lysimeters can be used for research.

References

- Baumann, K. (2020). Phosphorus cycling and spring barley crop response to varying redox potential. *Vadose Zone Journal*. University of Rostock. https://www.researchgate.net/publication/347834570_Phosphorus_cycling_and_spring_barley_crop_response_to_varying_redox_potential/link/6478cd8cb3dfd73b7758c10b/download

- Baumann, K. (2022). Lysimeter Bedienungsanleitung. University of Rostock

- UGT. (2018). Tensio 160 *Datasheet*
- UGT. (2018). SMT100 *Datasheet*
- ecoTech. (2016). pH Elektroden *Datasheet*. *https://www.vanwalt.com/pdf/information-sheets/ecoTech-Manual-pH-electrodes.pdf*
- UGT. (2016). Redox Elektroden *Datasheet*. *https://www.vanwalt.com/pdf/information-sheets/ecoTech-Manual-Redox-electrodes.pdf*
- Thermo Fisher. (2017). DT80 Range User‘s Manual. https://www.manualpdf.in/datataker/dt-85/manual
- Thermo Fisher. (2020). Software, Firmware, and Drivers. https://www.thermofisher.com/de/de/home/industrial/manufacturing-processing/data-acquisition/software-firmware-drivers.html

Glossary

- IP address[1] : **Internet Protocol address** is a numerical label that is assigned to a device connected to a computer network that uses the Internet Protocol for communication.
- TCP/IP[2] : **Transmission Control Protocol/Internet Protocol** is the fundamental communication model of the Internet. It defines the set of rules that enable computers to interconnect, transfer data and communicate in a standardized way.
- FTP Server[3] : Software that enables file transfer over the Internet. It allows files to be sent from a client device to a server and downloaded from the server to the client device using **File Transfer Protocol.**
- RAM[4] **: Random-access memory** is a form of electronic computer memory that can be read and changed in any order, typically used to store working data.

Appendix

Appendix 1 : Monoliths 2
Appendix 2 : DT80 electronic Installation 2
Appendix 3 : Diagram of the Job - Full version 2
Appendix 4 : The whole programme 3
Appendix 5 : Initialization main program - Disables all items 4
Appendix 6 : Initialization of main program - waits for the user to enter the three information items 4
Appendix 7 : Initialization main program - Trying to communicate with the DT80 4
Appendix 8 : Initialization main program - Unlocking every items if the communication work 4
Appendix 9 : Main program - ON/OFF water regulation 5
Appendix 10 : Main program - Water Level control 5
Appendix 11 : Main program - Measurement frequency control 6
Appendix 12 : Main program - Sensor calibration 6
Appendix 13 : Log generator - measurement frequenc 7
Appendix 14 : Log generto- water level 7
Appendix 15 : Log generator - Sensors calibration 8
Appendix 16 : Timing security program 8

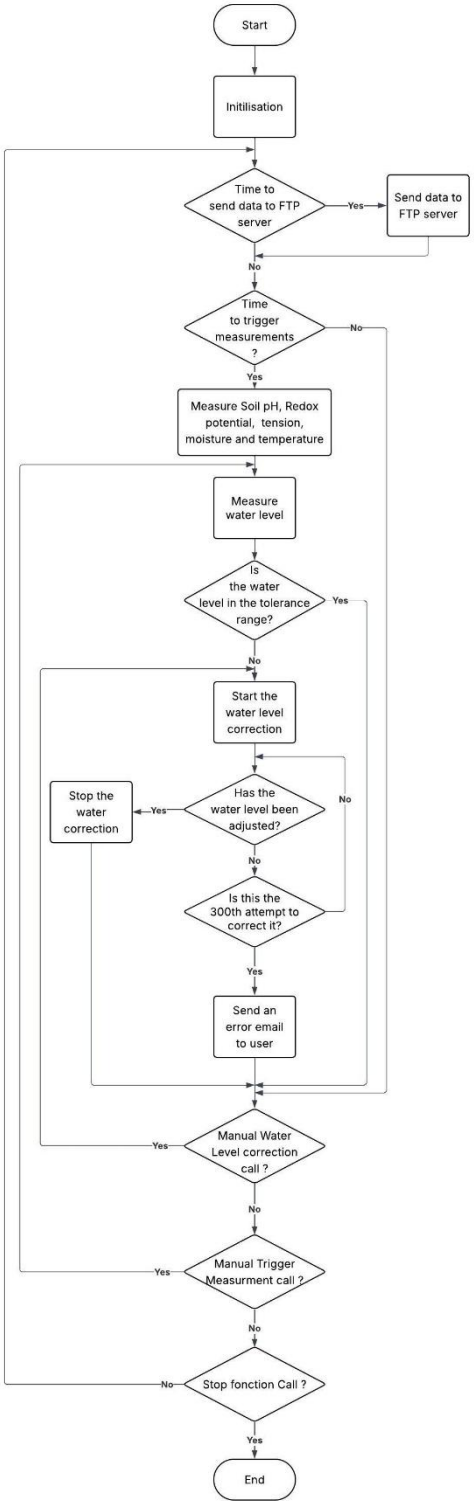
1. Device



Appendix 1 : Monoliths

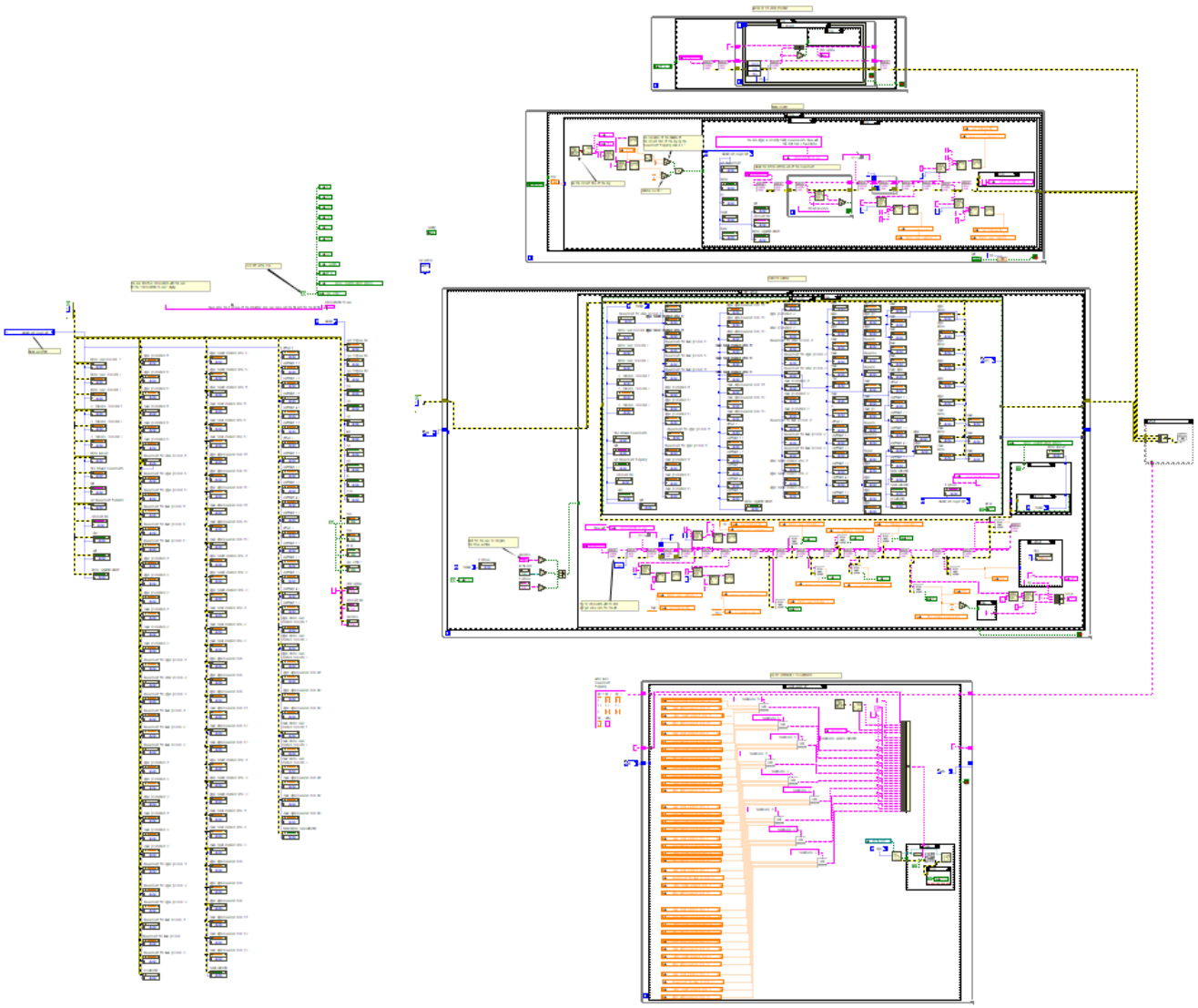


Appendix 2 : DT80 electronic Installation



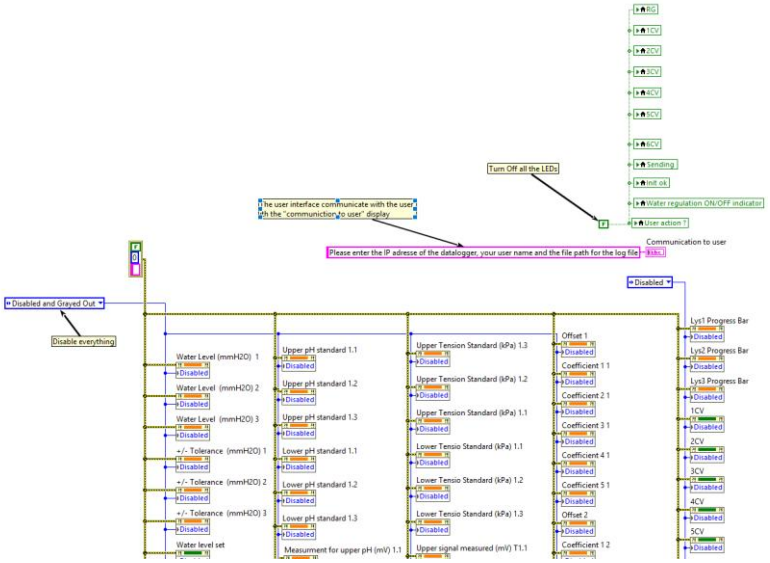
Appendix 3 : Diagram of the Job - Full version

2. LabVIEW program

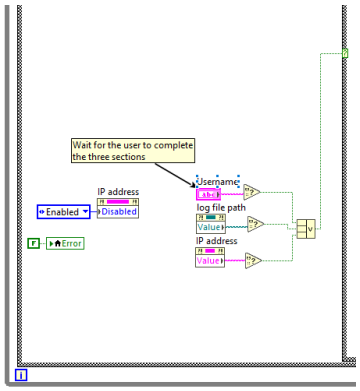


Appendix 4 : The whole programme

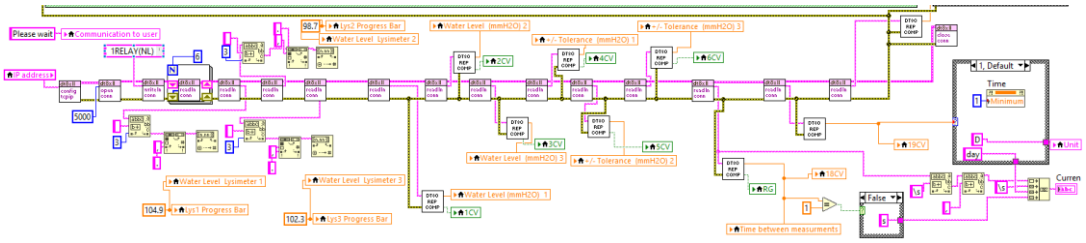
2.1. Main program



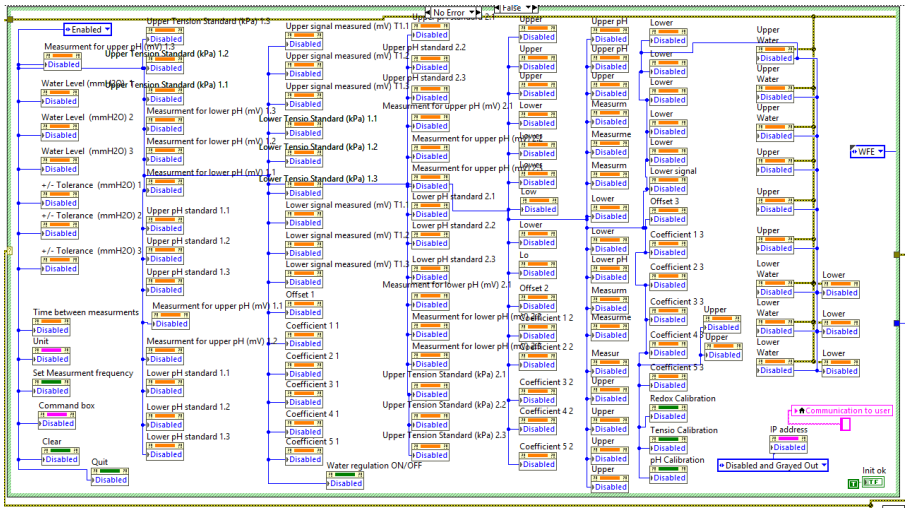
Appendix 5 : Initialization main program - Disables all items



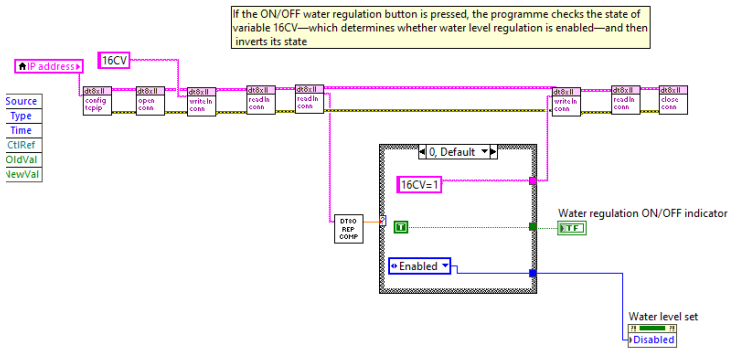
Appendix 6 : Initialization of main program - waits for the user to enter the three information items



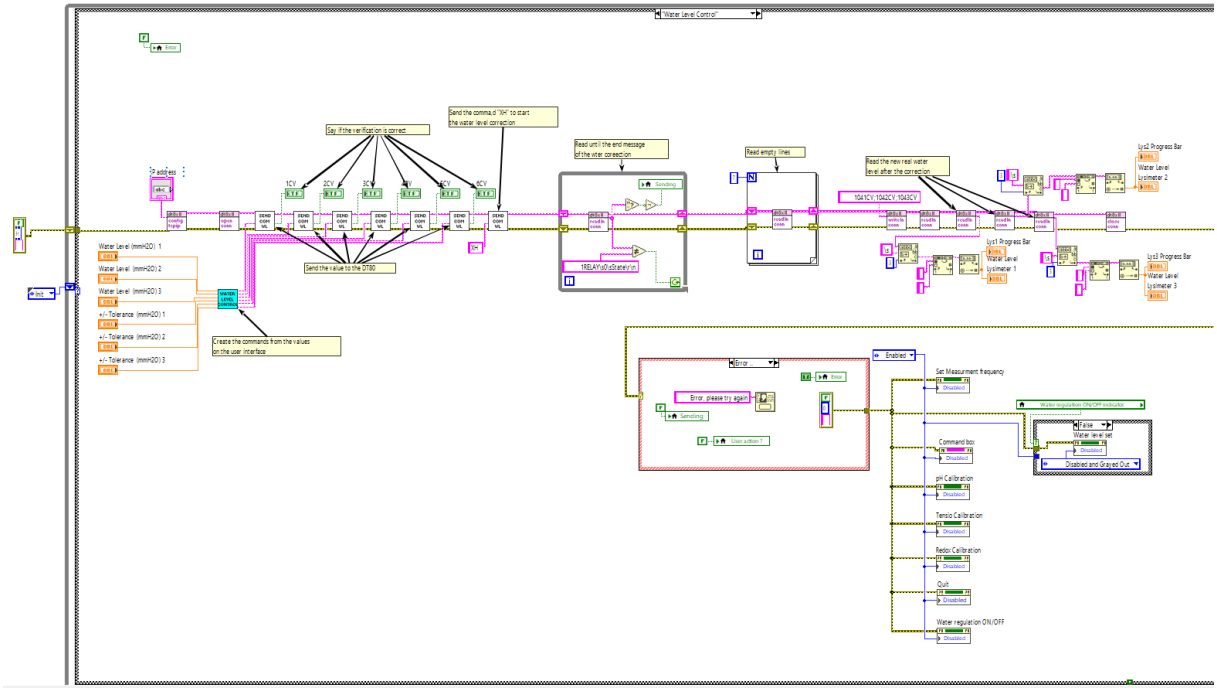
Appendix 7 : Initialization main program - Trying to communicate with the DT80



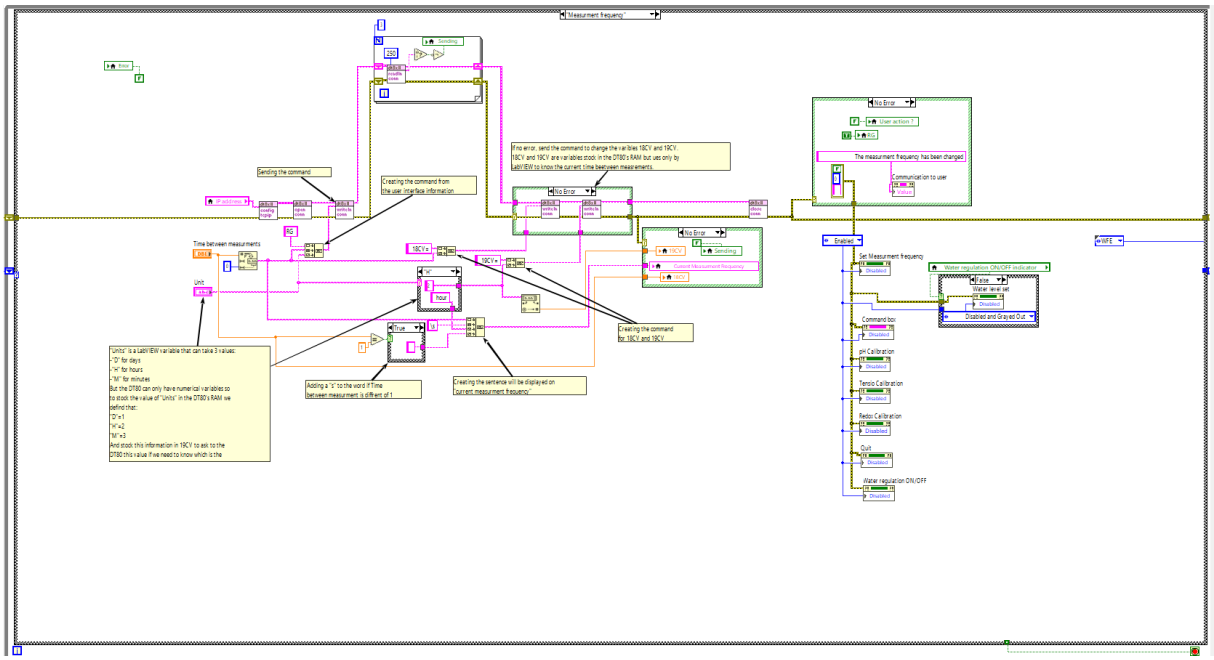
Appendix 8 : Initialization main program - Unlocking every items if the communication work



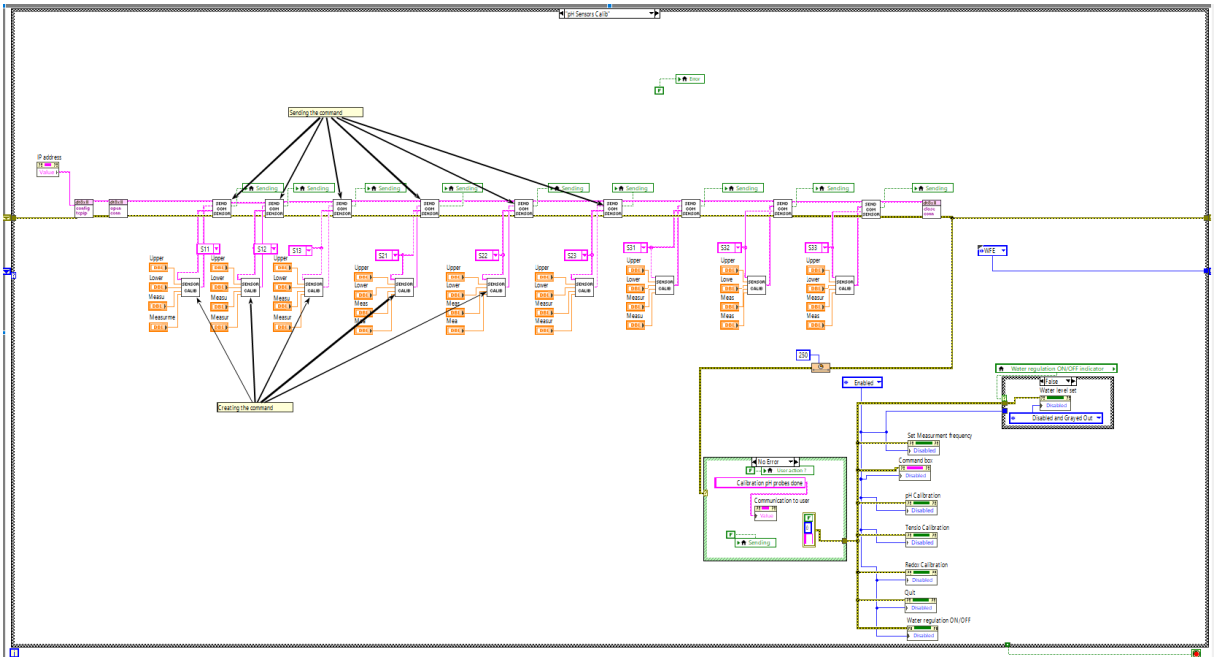
Appendix 9 : Main program - ON/OFF water regulation



Appendix 10 : Main program - Water Level control

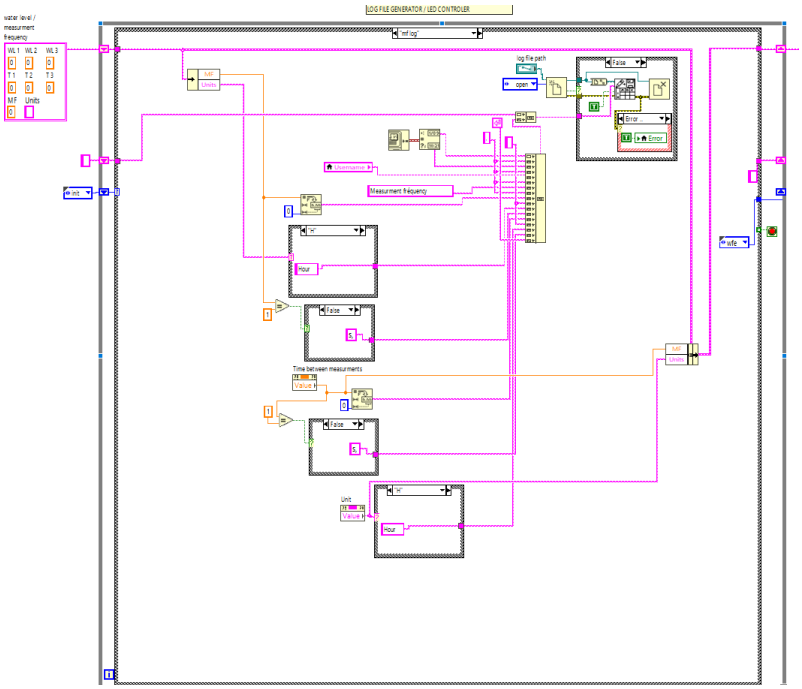


Appendix 11 : Main program - Measurement frequency control

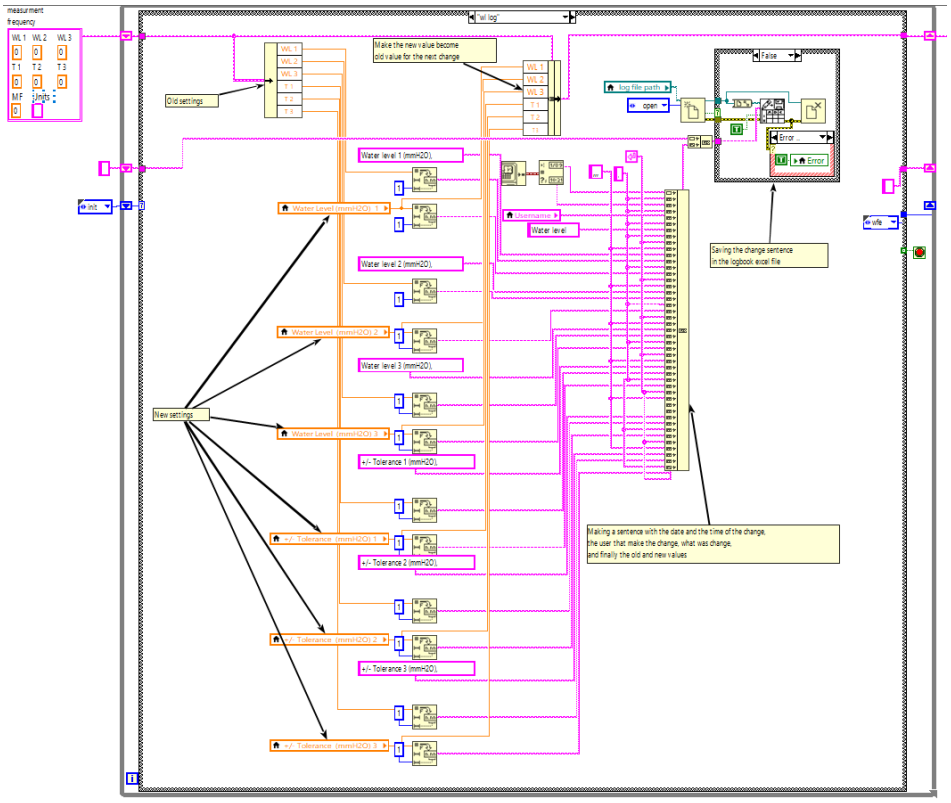


Appendix 12 : Main program - Sensor calibration

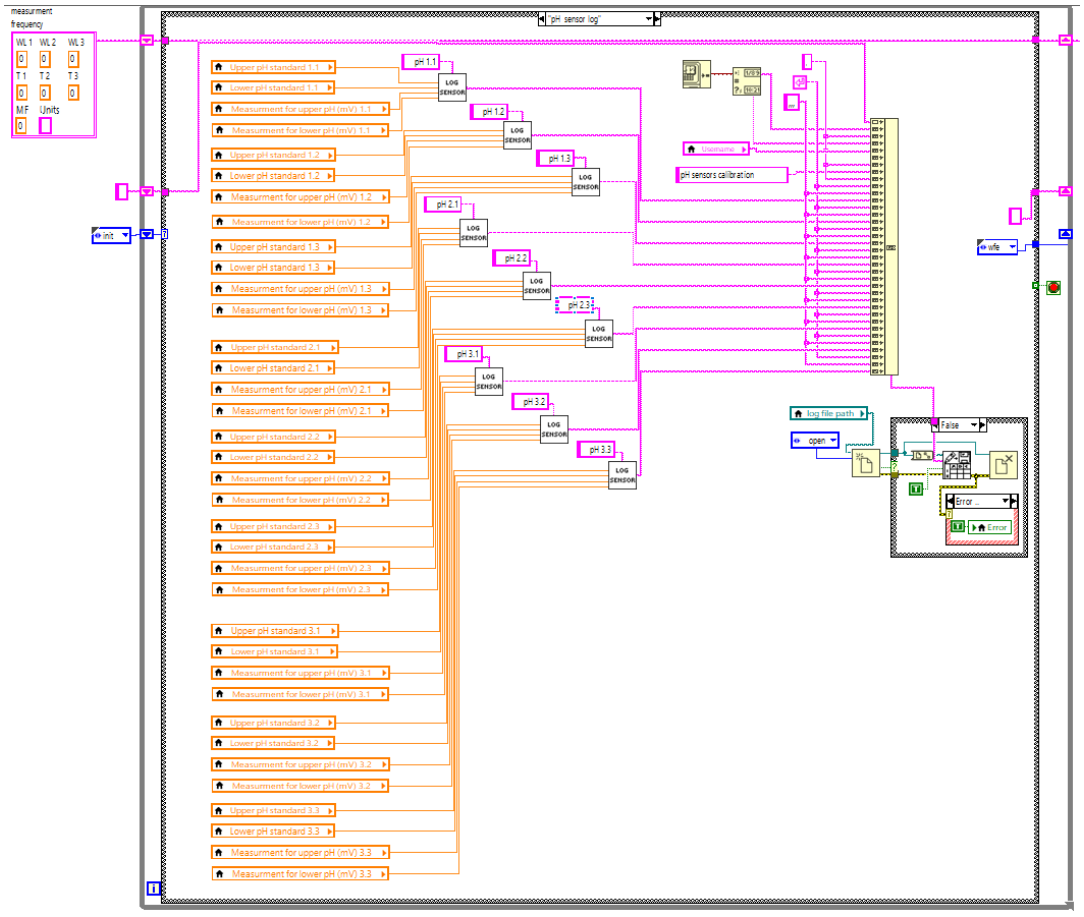
2.2. Log generator program



Appendix 13 : Log generator - measurement frequenc

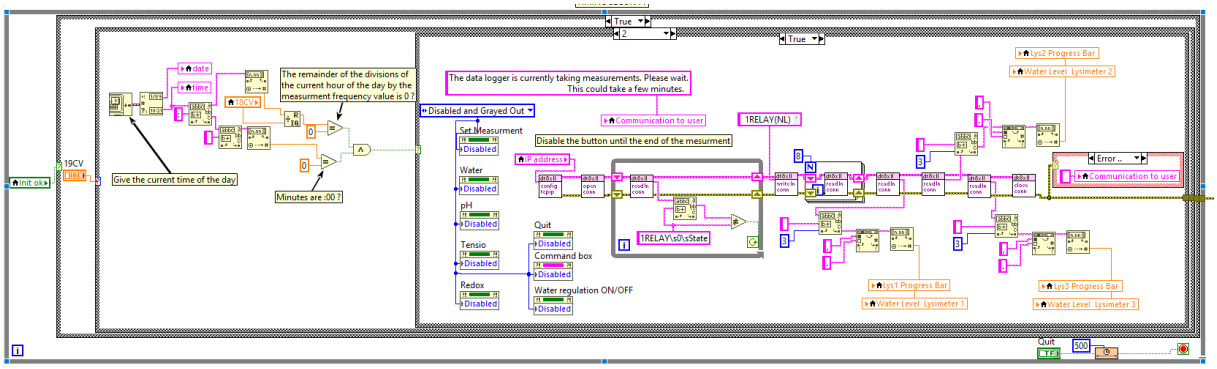


Appendix 14 : Log generto- water level



Appendix 15 : Log generator - Sensors calibration

2.3. Timing security



Appendix 16 : Timing security program